

QUANTIUM DATA ANALYTICS JOB SIMULATION

Retail Analytics: Customer Segments & Store Trial Analysis

1. Executive Summary

This project analyzes customer purchasing behaviour in the chips category and evaluates the performance of a store trial conducted across stores **77, 86, and 88**.

The analysis was conducted using transaction and customer data to identify important customer segments, purchasing patterns, and differences in spending behaviour. A second phase evaluated whether the store trial generated measurable sales improvements by comparing each trial store with a matched control store.

The customer analysis identified **Older Families and Young Families** as the highest-spending customer segments, with average spend of approximately **\$32–\$35 per customer**. Young Singles/Couples represented the largest customer group, with **7,917 Mainstream shoppers**, but had the lowest average spend per customer at approximately **\$18.64**.

The store trial analysis showed substantial sales uplift during the trial period, with the highest monthly uplift reaching **62.3% at Store 77 in April 2019**. The analysis indicates that increased customer numbers were the primary driver of the improvement rather than larger baskets.

2. Business Problem

The business wanted to understand two key areas:

Customer behaviour

- Which customer segments generate the most value?
- Which segments represent the largest customer base?
- How much does each customer segment spend?
- Where are there opportunities to increase customer spending?

Store trial performance

- Did the trial stores experience increased sales?
- Was the sales change statistically significant?
- Which stores showed the strongest uplift?
- Was growth driven by acquiring more customers or increasing spending from existing customers?

The analysis was designed to provide data-driven insights that could support customer targeting and store-level decision-making.

3. Dataset & Data Preparation

The analysis used transaction-level retail data together with customer purchasing behaviour data.

Key data fields included

Transaction data

- Transaction date
- Store information
- Product information
- Product quantity
- Product price
- Loyalty card number

Customer data

- Loyalty card number
- Lifestage
- Premium customer segment

The data was loaded and processed using **Python and Pandas**.

Data preparation steps

The following activities were performed:

1. Inspected dataset structure and data types.
2. Converted transaction dates into appropriate datetime format.
3. Examined product names and removed salsa products from the chips analysis.
4. Investigated product quantity outliers.
5. Identified two transactions containing an abnormal quantity of **200 units** and removed them.
6. Extracted product pack sizes from product names.
7. Extracted and standardised brand names.
8. Merged transaction data with customer information.
9. Checked the resulting dataset for missing values and data quality issues.

After cleaning, the transaction dataset contained **246,740 records** for analysis.

4. Customer Segmentation Analysis

Customer segments were analysed using **lifestage** and **premium customer** classifications.

The analysis examined:

- Total sales
- Number of customers
- Average spend per customer
- Average pack size
- Customer segment behaviour

Average Spend by Segment

Customer Segment	Average Spend
Older Families	\$34.58
Young Families	\$32.82
Older Singles/Couples	\$26.39
Midage Singles/Couples	\$25.69
Retirees	\$24.15
Young Singles/Couples	\$18.64

The values show a clear difference in spending behaviour between customer groups.

Key Finding

Older Families and Young Families recorded the highest average spend per customer, making these segments important from a customer-value perspective.

At the same time, **Young Singles/Couples** represented the largest customer group, with **7,917 Mainstream shoppers**, but their average spend per customer was substantially lower.

This creates two distinct strategic opportunities:

- **Families** → retain existing high-value customers
- **Young Singles/Couples** → increase spend per visit

5. Store Trial Analysis

Three stores were selected as trial stores:

- **Store 77**
- **Store 86**
- **Store 88**

Matched control stores were used to estimate how the trial stores would have performed without the intervention.

Matched Controls

Trial Store	Control Store
Store 77	233
Store 86	155
Store 88	203

The analysis compared trial-store performance against the corresponding control stores during the trial period.

6. Trial Period Results

The trial period covered:

February 2019 – April 2019

Sales uplift compared with control

Store	Feb 2019	Mar 2019	Apr 2019
Store 77	-5.9%	+36.7%	+62.3%
Store 86	+5.6%	+31.6%	+3.5%
Store 88	+16.3%	+20.9%	+39.6%

These results are presented in the project analysis.

Statistical Significance

Statistical testing was performed using a **95% confidence level**.

The analysis identified significant sales uplift during the trial period, with:

- Store 77 showing significant uplift in March and April.
- Store 86 showing significant uplift in March.
- Store 88 showing significant uplift across the trial months.

Store 88's matched control had a weaker correlation than the controls used for the other stores, so its results should be interpreted with additional caution.

7. What Drove the Sales Growth?

A key objective was to determine whether sales growth came from:

1. More customers visiting the stores, or
2. Existing customers spending more.

The analysis found that **customer acquisition was the primary driver**.

Customer counts increased substantially during the trial, while the change in spend per customer was comparatively small.

Average uplift by driver

Store	More Customers	More Transactions per Customer
Store 77	26.8%	-2.0%
Store 86	13.7%	0.7%
Store 88	14.3%	8.3%

The analysis therefore indicates that the trial's main effect was **bringing more shoppers into the stores rather than substantially increasing basket size**.

8. Key Insights

Insight 1 — Families are high-value customers

Older and Young Families recorded the highest average spending, approximately **\$32–\$35 per customer**.

Insight 2 — Young Singles/Couples represent a large opportunity

Young Singles/Couples were the largest customer group but had relatively low spending per visit. This indicates an opportunity to increase basket size.

Insight 3 — Store 77 recorded the largest monthly uplift

Store 77 reached **62.3% sales uplift in April 2019**, the highest monthly uplift observed across the three trial stores.

Insight 4 — Customer acquisition drove trial performance

The analysis indicates that the increase in sales was primarily associated with **more customers**, rather than substantially higher spending per customer.

Insight 5 — Store 88 requires additional validation

Store 88's matched control had a weaker pre-trial relationship than the other control matches, with a reported correlation of **0.51 versus 0.9+** for the other stores.

9. Recommendations

Based on the analysis:

1. Expand the tested store initiative

The trial generated measurable sales improvements across the tested stores, with customer growth being the main driver. The analysis therefore supports extending the tested initiative to a wider set of stores, while continuing to monitor performance.

2. Increase basket size among Young Singles/Couples

Young Singles/Couples represent a large customer base but have relatively low spending per visit.

Potential approaches include:

- Multi-buy offers , Larger pack promotions
- Targeted product combinations
- Offers designed to encourage additional purchases

3. Retain high-value Family customers

Older and Young Families already generate higher average spending.

Retention-focused approaches could include:

- Loyalty offers
- Family-oriented promotions
- Repeat-purchase incentives
- Larger-pack offers

These recommendations align with the final presentation's proposed next steps.

10. Tools & Technologies

Programming & Analysis

- Python
- Pandas , NumPy

Visualization

- Matplotlib
- Power BI

Statistical Analysis

- Correlation analysis
- Control-store matching
- Percentage uplift analysis
- Statistical significance testing
- 95% confidence level

Data Skills

- Data cleaning
- Data transformation
- Exploratory data analysis
- Customer segmentation
- Trend analysis , Business insight generation

11. Conclusion

This project demonstrated an end-to-end retail analytics workflow, from data preparation and customer segmentation through to statistical evaluation of a store trial.

The analysis identified clear differences in customer value across segments and found that the store trial's sales improvement was primarily associated with **increased customer acquisition rather than larger customer baskets**.

The results provide a data-driven basis for customer targeting, retention strategies, basket-growth initiatives, and evaluation of future store-level trials.